

Surface Texture Analysis of Machined Components

Measuring roughness across machining processes and evaluating vibratory finishing

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PROJECT SUMMARY

This project examined how machining and finishing methods affect surface texture. Surface roughness (Ra and Rq) was measured on lathe-turned, milled, superfinished, and CNC-machined specimens using a microfinish comparator, a perthometer, and a stylus-based surfometer. A CNC part machined at three feed rates was then polished in a vibratory finisher and re-measured. The superfinished specimen was by far the smoothest ($Ra \approx 2.7 \mu\text{in}$), the turned rod regions varied widely ($15\text{--}63 \mu\text{in}$), and vibratory finishing reduced roughness at every feed rate — by as much as 43% on the part milled at 25 in/min. Slower feed rates consistently produced smoother surfaces.

1. Introduction & Objectives

This project explored the link between processing parameters and surface accuracy. Surface finish — including roughness, waviness, and lay — is a key indicator of how well a machined component will perform. The objectives were to determine how different manufacturing and finishing techniques affect surface texture; become familiar with the measuring instruments and comparators used to quantify it; learn the standard symbols for surface roughness; and evaluate the effect of vibratory finishing.

2. Background

Surface texture is shaped by the manufacturing process, so engineers need to understand how each method affects the final finish. Texture is quantified with stylus-based instruments or optical profilers for high-precision results. Finishing processes such as barrel tumbling and vibratory finishing are widely used to clean and smooth surfaces after machining; their speed and effectiveness depend on the material, the media, and the technique used.

3. Procedure

With the TA's guidance, the team measured the roughness of a plastic specimen, a milled metal specimen, four regions of a turned rod, and a CNC-machined plate milled at feed rates of 50, 25, and 10 in/min. Each surface was first graded by touch and sight against a microfinish comparator, then measured with the perthometer and surfometer for more accurate data. The CNC plate was then placed in a water-filled vibratory finisher for 45 minutes and re-measured. Results and standard roughness symbols were recorded on the data sheet.

4. Results

Table 1. Surface finish measurements by instrument

Specimen	Comparator Ra (μin)	Perthometer Ra (μin)	Perthometer Ra (μm)	Surfometer Ra (μin)	Surfometer Ra (μm)	Likely process
Plastic	32	2.656	0.0675	8.0	0.2032	Superfinishing
Metal	63	49.686	1.2620	166.0	4.2164	Milling
Rod, region 1	32	15.012	0.3813	35.4	0.8992	Turning
Rod, region 2	63	48.599	1.2344	78.4	1.9914	Turning
Rod, region 3	16	16.897	0.4292	23.4	0.5944	Turning
Rod, region 4	125	62.625	1.5907	74.0	1.8796	Turning

Table 2. CNC part roughness before and after 45 minutes of vibratory finishing (perthometer)

Feed (in/min)	As machined Rq (μin)	As machined Ra (μin)	After finishing Rq (μin)	After finishing Ra (μin)	Ra reduction
50	227.388	183.841	220.780	173.663	5.5%
25	153.285	125.412	81.982	71.173	43.2%
10	48.959	40.272	37.968	32.297	19.8%

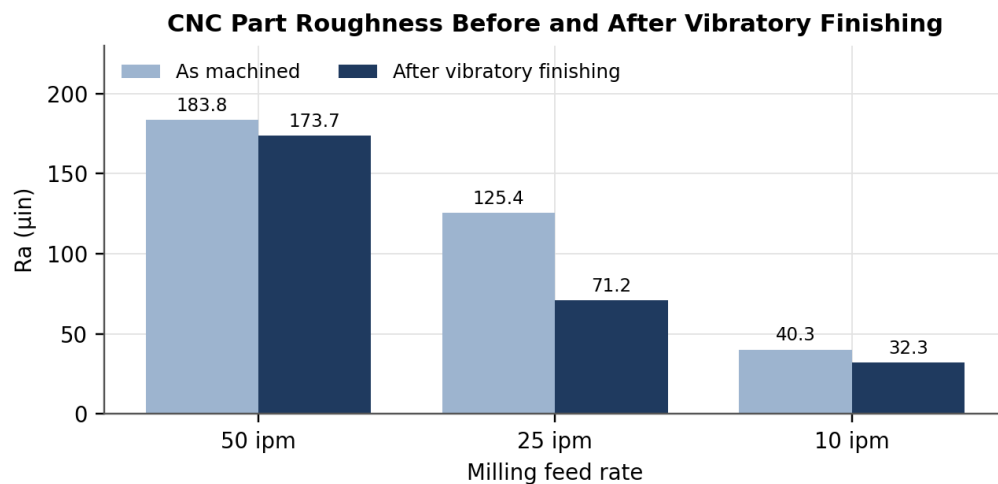


Figure 1. Ra of the CNC part at each feed rate, before and after vibratory finishing.

5. Discussion

Turning. The four regions of the turned rod ranged from about 15 to 63 μin on the perthometer, and region 4 was the roughest on every instrument (125 μin on the comparator). That spread on a single part points to changes in cutting conditions and tool condition along the rod, and shows how strongly cutting speed, feed rate, and tool quality influence surface finish. Region 4 was rougher than expected for a turned surface; tool wear, variation in feed or cutting speed, material properties affecting machinability, or coolant problems are all possible causes.

Superfinishing vs. milling. The superfinished plastic specimen was the smoothest surface measured ($R_a = 2.66 \mu\text{in}$), supporting the idea that superfinishing greatly improves surface quality. The milled metal specimen was noticeably rougher ($R_a = 49.7 \mu\text{in}$), which is typical of milling. Together these results show why selecting the right finishing method matters for achieving the required surface.

Feed rate and vibratory finishing. On the CNC part, roughness dropped sharply with feed: 183.8 μin at 50 in/min versus 40.3 μin at 10 in/min. Vibratory finishing reduced roughness at all three feed rates, although the improvement depended on the starting surface — only about 6% on the roughest surface, but 43% at 25 in/min and 20% at 10 in/min. This consistent trend shows the value of vibratory finishing for refining surfaces after machining.

Sources of error. The measurement patterns were consistent, but the variation between specimens suggests some inconsistency in process control. Sample placement during measurement, surface cleanliness, and instrument calibration are the most likely sources of error. The three instruments also differ in method — the comparator is a subjective visual and tactile grade — so absolute values vary between them, while trends agree.

6. Conclusion

The results confirmed that machining and finishing methods directly determine surface texture. Turned surfaces varied with cutting conditions and tool wear, milling produced a characteristically rougher surface, and superfinishing produced the smoothest surface by a wide margin. Lower feed rates produced smoother milled surfaces, and vibratory finishing lowered roughness at every feed rate tested. Overall, the project showed how important process control and the right finishing technique are for achieving a required surface quality.

7. Key Concepts

Which instruments measure surface roughness?

Microfinish comparators (visual and tactile reference standards), perthometers, and stylus-based surfometers. Surface roughness can also be observed under optical microscopes and scanning electron microscopes.

How does feed rate affect roughness?

Slower feed rates produce lower roughness, as shown clearly by the CNC part in this project.

What does vibratory finishing do?

It produces smoother surfaces and reduces overall roughness by tumbling parts with abrasive media.

How does interferometry measure a surface?

It analyzes interference patterns created by light beams reflected from the surface; the displacement of those patterns is used to calculate surface height.

What does an angular lay symbol mean?

The dominant direction of the surface marks runs at an angle to the reference surface.

How do grinding and turning affect fatigue performance?

Both generate heat at the surface that can alter the material and lower the endurance limit of steel.

Which processes produce a given surface roughness range?

Common processes include milling, drawing, sawing, forging, turning, permanent mold casting, EDM, laser cutting, chemical milling, cold rolling, planing, shaping, drilling, boring, grinding, investment casting, extrusion, and electron beam cutting — each with its own typical range.